Question: Given the equation $x^2-(k+2)x+k+1=0$, find the real number k such that one root of the equation is twice the other.

Student Model

Self-generated Errors

Let's denote the roots by α and 2α. By Vieta's formulas, for the quadratic equation $x^2-(k+2)x+k+1=0$, the sum of the roots is given by $\alpha+2\alpha=3\alpha=k+2$ and the product of the roots is given by $\alpha\cdot 2\alpha=2\alpha^2=k+1$. First, solve for α in terms of k from the sum of the roots equation: $3\alpha=k+2\Rightarrow\alpha=\frac{k+2}{3}$. Next, substitute $\alpha=\frac{k+2}{3}$ into the product of the roots equation: $2\left(\frac{k+2}{3}\right)^2=k+1$. Expanding and simplifying, we get: $2k^2+4k+4=9k+9$. Moving all the terms to one side of the equation: $2k^2-5k-5=0$. This is a quadratic equation in k. We can solve it using the quadratic formula .... So, $k=\frac{5\pm\sqrt{65}}{4}$.

Teacher Model

Deliberately Introduced Errors

The given quadratic equation is: $x^2-(k+2)x+(k+1)=0$.
We can factorize it as: $(x-1)(x-(k+1))=0$. This gives us the roots: $x=1$ and $x=k+1$. Since one root is twice the other, we can set up the equation: $1=2(k+1)$. Solving for k, we get: $k=-\frac{1}{2}$. So, the value of k is $-\frac{1}{2}$.

Flawed Trajectory Pool

Fix & Continue Revision

To find the value of k such that one root of the quadratic equation $x^2-(k+2)x+k+1=0$ is twice the other, we can ... substitute $\alpha=\frac{k+2}{3}$ into the product of the roots equation: $2\left(\frac{k+2}{3}\right)^2=k+1$. Expanding and simplifying, we get: $2k^2+4k+4=9k+9$. Wait, this doesn't seem right. Let's re-examine the calculation step by step. The correct expansion of $2\left(\frac{k+2}{3}\right)^2$ is: $2\cdot\frac{(k+2)^2}{9}=\frac{2k^2+8k+8}{9}$. So, the equation becomes: $\frac{2k^2+8k+8}{9}=k+1$. Multiply both sides by 9: $2k^2+8k+8=9k+9$. Simplify: $2k^2-k-1=0$. Now, solve this quadratic equation for k: ... So, the possible values of k are: 1 and $-\frac{1}{2}$.

Fresh & Restart Revision

To find the value of k ... By Vieta's formulas, ... Next, substitute $\alpha=\frac{k+2}{3}$ into the product of the roots equation: $2\left(\frac{k+2}{3}\right)^2=k+1$. Expanding and simplifying, we get: $2k^2+4k+4=9k+9$. Hmm, this seems complex. Is there a smarter way to solve it? I notice that the given equation $x^2-(k+2)x+k+1=0$ can be factorized as: $(x-1)(x-(k+1))=0$. This gives us the roots: $x=1$ and $x=k+1$. Since one root is twice the other, we have two cases to consider:
Case 1: $k+1=2\times 1$, solving for k, $k+1=2\implies k=1$. **Case 2:** $1=2\times(k+1)$, solving for k, $1=2k+2\implies 2k=-1\implies k=-\frac{1}{2}$. Thus, the possible values of k are 1and $-\frac{1}{2}$.

Revised Trajectory Pool

Smooth

Training Corpus